

PROVISIONAL PATENT APPLICATION

**STRUCTURAL INTEGRITY MEASUREMENT BY
CONTROLLED REPRESENTATION DISASSEMBLY
AND RECONSTRUCTION SYSTEMS AND METHODS**

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Abstract

Systems and computer-implemented methods are disclosed for measuring structural integrity of a computational representation by controlled disassembly, perturbation, isolation, compression, reconstruction, or substitution of representation components and by measuring resulting changes in organizational structure. A baseline representation is constructed from observations or computational states. One or more controlled operators selectively remove, suppress, shuffle, perturb, isolate, replace, compress, orthogonalize, residualize, or reconstruct selected representation components, including dimensions, feature families, relationships, paths, scales, hierarchy levels, neighborhoods, regions, or generators. Structural properties of the baseline and modified representations are measured and compared, including neighborhood preservation, rank preservation, connectivity, reachability, basin agreement, path or transport preservation, bottleneck preservation, recursive consistency, regime preservation, or scale stability. Structural damage may be normalized by the amount of representational content removed or altered to detect components whose organizational importance is disproportionate to their coordinate variance, dimensionality, frequency, or other conventional importance measures. Integrity measurements may be maintained as a multidimensional vector rather than a single scalar and may be used to select representations, identify distributed structural dependence, trigger re-indexing or replacement, limit downstream use, or request additional evidence. The methods may operate independently of supervised labels and may be applied to learned embeddings, recursive representations, graphs, dynamical systems, biological representations, world models, scientific data, or other structured computational systems.

1. Technical Field

The present disclosure relates generally to computer-implemented representation analysis, structural diagnostics, graph and topological computation, recursive indexing, manifold analysis, model monitoring, dimensionality analysis, robustness testing, scientific computing, machine learning, and representation engineering.

More particularly, the disclosure relates to systems and methods for determining whether a computational representation preserves important organizational structure when selected portions of the representation are systematically removed, suppressed, isolated, perturbed, shuffled, compressed, reconstructed, or otherwise modified under controlled conditions.

2. Background

Computational representations are commonly evaluated using dimensionality, reconstruction error, explained variance, predictive performance, embedding distance, calibration, or task-specific accuracy. These measurements can be useful, but they do not necessarily establish whether the relational organization of a representation depends on components that appear weak under ordinary coordinate-centric measures.

A component can account for little coordinate variance yet still materially affect neighborhood identity, path accessibility, connectivity, bottleneck structure, recursive hierarchy, or regime boundaries. Conversely, a component can account for substantial variance while contributing little to a selected structural relationship. The importance of a representation component therefore cannot always be inferred from variance magnitude, feature weight, predictive coefficient, or individual ablation score alone.

Conventional ablation studies often remove model modules, features, neurons, or inputs and measure downstream task performance. Such studies usually answer whether a component affects a task outcome. They do not necessarily measure whether the representation itself remains organizationally intact. In scientific or safety-sensitive settings, task labels may be unavailable, delayed, contaminated by tuning, or inappropriate for determining intrinsic representation quality.

Topological and graph-robustness methods measure useful properties of networks and manifolds, but a need remains for a general representation-integrity architecture in which controlled disassembly is defined over multiple component types and scales, structural preservation is measured relative to a frozen baseline, damage is normalized by the amount or type of representational content altered, and the results are used to characterize distributed versus localized structural dependence.

The inventors recognized that representation quality can be studied by treating the representation as an object that may be deliberately disassembled and then measuring which organizational properties survive. This permits structural importance to be separated from coordinate magnitude and permits intrinsic, phenotype-blind integrity analysis before downstream prediction or scientific interpretation.

3. Summary of the Disclosure

In one aspect, a baseline representation is constructed and one or more baseline structural properties are measured. A controlled disassembly operator modifies selected representation components while preserving other components or construction rules. The modified representation is then structurally re-evaluated and compared with the baseline.

In another aspect, the disassembly operator removes or suppresses one or more coordinate dimensions, principal components, latent factors, feature families, event families, structural channels, graph edges, paths, hierarchy levels, recursive scales, regions, landmarks, or generator components.

In another aspect, the operator does not remove a component but instead shuffles, permutes, randomizes, substitutes, orthogonalizes, residualizes, compresses, reconstructs, or replaces the component with a matched control so as to distinguish structural dependence from simple dimensionality or capacity effects.

In another aspect, structural damage is measured using one or more properties including neighborhood overlap, neighbor-rank correlation, graph connectivity, edge-rank preservation, reachability, basin or region agreement, path accessibility, path-length distortion, transport robustness, bottleneck preservation, recursive consistency, regime preservation, or structural recovery after reconstruction.

In another aspect, the structural damage is normalized by a quantity representing removed or altered representational content. Such quantity may include variance removed, dimensions removed, feature mass, graph edge mass, occupancy mass, information measure, path mass, computational cost, or another measure. The normalized result identifies components producing disproportionate structural damage.

In another aspect, multiple disassembly scales or component blocks are evaluated to generate an integrity profile or integrity surface rather than a single score. The profile can reveal whether organization is concentrated, distributed, redundant, multiscale, fragile, or resilient.

In another aspect, the system distinguishes controlled disassembly from scientific qualification. Integrity measurements describe how a representation behaves under structural perturbation. A separate subsystem may later determine whether such evidence supports scientific interpretation, prediction, autonomous action, regulatory use, or other downstream conclusions.

In another aspect, the system operates without supervised target labels. Structural integrity can therefore be measured before outcome labels are accessed, reducing leakage between representation construction and scientific or predictive evaluation.

In another aspect, integrity measurements may be used by a representation runtime or engineering system to choose among candidate representations, request reconstruction, broaden recursive scale, materialize additional hierarchy, switch to an alternate representation, or invalidate a structural view for selected operations.

4. Brief Description of the Drawings

FIG. 1. Overall integrity-analysis architecture from baseline representation through controlled disassembly, structural remeasurement, normalization, and integrity profile generation.

FIG. 2. Example taxonomy of controllable representation components including dimensions, feature families, relationships, paths, regions, scales, hierarchy levels, and generators.

FIG. 3. Example controlled-disassembly operators including removal, masking, shuffling, substitution, compression, orthogonalization, residualization, isolation, and reconstruction.

FIG. 4. Example baseline-versus-modified neighborhood comparison.

FIG. 5. Example transport and path preservation analysis.

FIG. 6. Example normalization of structural damage by removed representational content.

FIG. 7. Example integrity profile across multiple disassembly scales.

FIG. 8. Example distributed-structure condition in which both leading and trailing representation components contribute to local organization.

FIG. 9. Example reconstruction/recovery integrity test.

FIG. 10. Example interaction with a representation runtime, structural engine, and independent qualification subsystem.

FIG. 11. Example phenotype-blind experimental protocol separating representation integrity from later outcome evaluation.

FIG. 12. Example cross-representation comparison using matched disassembly operators.

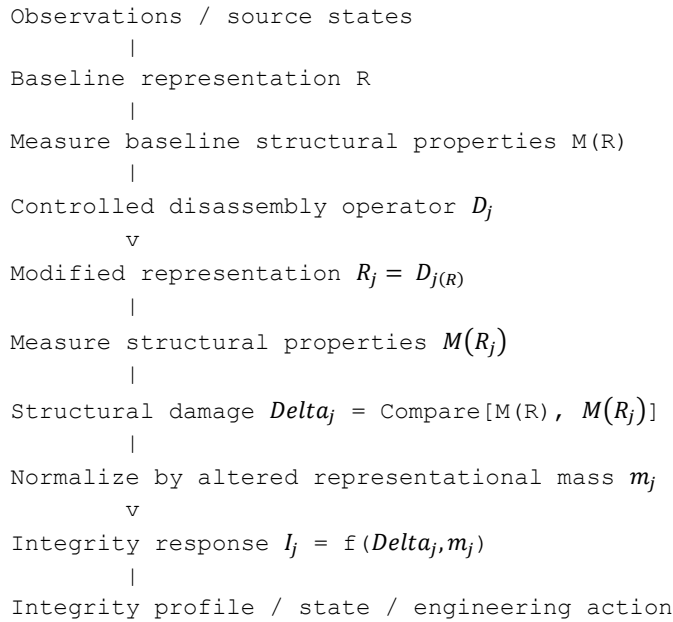
5. Definitions and Interpretive Conventions

Term	Meaning
Representation	Any computational organization of observations, states, coordinates, embeddings, identities, relationships, constraints, paths, graphs, neighborhoods, regions, hierarchy, topology, or derived structural objects.
Baseline representation	A representation selected as the reference state for an integrity analysis before application of a controlled modification.
Controlled disassembly	A deliberate, parameterized modification of one or more representation components for the purpose of measuring structural dependence. Disassembly may be destructive or reversible and does not require physical deletion.
Controlled reconstruction	Reintroduction, estimation, substitution, regeneration, or restoration of a removed or altered component followed by structural remeasurement.

Component	Any defined part of a representation, including dimensions, principal components, features, feature families, relations, edges, paths, neighborhoods, regions, basins, scales, hierarchy levels, generator terms, landmarks, channels, or other structural blocks.
Structural property	A measurable organizational property of the representation, including neighborhood identity, neighbor ordering, connectivity, reachability, basin membership, path accessibility, path length, transport, bottlenecks, recursion, hierarchy, regime, stability, or another relational or topological measurement.
Structural damage	A difference, divergence, loss, distortion, instability, or degradation in one or more structural properties between a baseline and modified representation.
Representational mass	A quantity used to characterize the amount of representational content altered by a disassembly. It may be measured using variance, number of dimensions, norm, feature energy, edge count, path mass, occupancy, information measure, computational burden, or another quantity.
Integrity	The degree to which organizational structure is preserved under one or more controlled modifications of a representation.
Distributed structural dependence	A condition in which organizational behavior depends materially on multiple components or scales rather than being recoverable from one dominant component or compact subset.
Redundant structural dependence	A condition in which one or more components can be removed with limited structural damage because other components preserve similar organization.
Disproportionate structural influence	A condition in which structural damage caused by removing or altering a component is large relative to the component's representational mass under a selected normalization.
Phenotype-blind or target-blind	Computed without accessing a downstream supervised target, outcome, phenotype, reward, decision label, or other task-specific response used

for later predictive evaluation.

6. Overall Integrity Analysis Architecture



The baseline and modified representations may use the same structural engine, different structural engines, or frozen structural-computation parameters. Preferably, parameters unrelated to the selected disassembly are held constant so that the observed structural response can be attributed to the controlled modification rather than to unrelated tuning.

7. Controlled Disassembly Operators

Operator class	Illustrative operation
Removal / masking	Remove, zero, mask, deactivate, or omit selected components.
Isolation	Retain a selected component or block while removing other components to determine whether the component alone preserves organization.
Shuffling / permutation	Randomize component-to-observation assignment, row order, edge assignment, path membership, or other correspondence while preserving one or more marginal distributions.
Substitution	Replace a component with a matched control, alternate generator output, baseline value, orthogonal component, randomized component, or independently constructed component.

Compression	Reduce dimensionality, aggregate structure, quantize, prune, coarsen, or otherwise reduce representational content.
Orthogonalization	Remove variation explainable by another component or subspace and evaluate the remaining residual organization.
Residualization	Construct a residual representation after regressing or projecting away a defined component or structural family.
Scale removal	Remove leading, trailing, intermediate, local, global, coarse, or fine structural scales.
Relationship/path removal	Remove edges, transformations, pathways, bottleneck regions, selected adjacency classes, or transport channels.
Hierarchy removal	Remove or virtualize one or more hierarchy levels or recursive abstraction layers.
Perturbation	Add noise, jitter, edge perturbation, coordinate perturbation, weight perturbation, or randomized rewiring within a controlled budget.
Reconstruction	Reintroduce a component using original, estimated, regenerated, imputed, compressed, or alternate content and measure structural recovery.

7.1 Matched-control operators

A preferred embodiment uses matched controls that preserve one or more nuisance properties while destroying a selected structural correspondence. Examples include shuffling a structural block while preserving its marginal distribution, replacing a component with a complexity-matched random component, or comparing a modified representation against a direct representation of equal dimensionality. Matched controls help distinguish structural dependence from generic capacity, dimensionality, or regularization effects.

8. Structural Preservation Measurements

Measurement	Non-limiting examples
Neighborhood preservation	Fraction or weighted fraction of baseline neighbors retained after modification.
Neighbor-rank preservation	Rank correlation or rank-distance agreement among local neighbors.

Edge-rank preservation	Agreement in ordering or weighting of representation edges.
Connectivity preservation	Agreement in component structure, reachability, giant-component size, or connectivity statistics.
Basin / region agreement	Agreement in region membership, boundary location, occupancy, purity, or transition relationships.
Path preservation	Agreement in accessible routes, shortest paths, path counts, path concentration, or route identity.
Transport preservation	Agreement in path lengths, transport costs, resistance-like quantities, transition accessibility, or robustness.
Bottleneck preservation	Agreement in critical edges, nodes, regions, or corridor dominance.
Recursive consistency	Agreement in higher-order structure across recursive indexing levels.
Scale stability	Consistency of selected structural properties across predefined scales.
Regime preservation	Agreement in identified structural regimes or regime boundaries.
Recovery after reconstruction	Degree to which structure returns when removed or altered components are restored or regenerated.

9. Integrity Metrics and Normalization

Let R denote a baseline representation and D_j a controlled disassembly operator. The modified representation is:

$$R_j = D_j(R)$$

For structural measurement M_k , a structural damage quantity may be defined generally as:

$$\Delta_{j,k} = d_k(M_k(R), M_k(R_j))$$

If m_j denotes the representational mass removed or altered by D_j , a normalized influence may be defined as:

$$S_{j,k} = \Delta_{j,k} / g(m_j)$$

where g may be identity, logarithmic, power-law, learned, bounded, or application-specific. A large S value can indicate disproportionate structural influence. Other embodiments may normalize by the fraction of remaining rather than removed content, or may jointly model several mass quantities.

A representation may be characterized by an IntegrityVector such as:

$$I(R) = [N, T, D, S]$$

- N: neighborhood-preservation integrity across predefined disassemblies.
- T: path/transport-preservation integrity across predefined disassemblies.
- D: decomposition sensitivity, including disproportionate structural damage relative to removed content.
- S: scale stability or multiscale consistency.

The vector is illustrative. Integrity need not be reduced to a scalar. Different applications may retain the full response surface indexed by operator, component, scale, and structural measurement.

10. Multiscale and Distributed Structural Dependence

A representation may contain structural information that is not concentrated in the components accounting for the most coordinate variance. The system therefore may apply complementary disassembly families across ordered components, such as leading versus trailing principal components, coarse versus fine hierarchy levels, local versus global graph scales, or high- versus low-frequency modes.

```
KEEP(q): retain a selected q-component subspace and remove the remainder
REMOVE(q): remove the selected q-component subspace and retain the remainder

Compare neighborhood / transport preservation for both families
across q in a predefined grid.
```

If removal of a high-mass leading subspace causes substantial local structural change and omission of a lower-mass trailing subspace also causes disproportionate local structural change, the representation may exhibit dual or distributed sensitivity. This condition can indicate that organizational structure depends on multiple scales or subspaces even when global variance appears concentrated.

10.1 Sensitivity normalized by removed or omitted mass

```
TailSensitivity(q) = [1 - Preservation_KEEP(q)] / OmittedMass(q)
HeadSensitivity(q) = [1 - Preservation_REMOVE(q)] / RemovedMass(q)
```

The naming is illustrative. The same architecture applies to feature families, graph partitions, pathway classes, or hierarchy levels.

11. Controlled Integrity Test Protocol

1. Freeze the source observations and baseline representation-construction procedure.
2. Construct the baseline representation without accessing downstream supervised targets when target-blind integrity is desired.
3. Freeze structural measurement definitions and disassembly operators.
4. Measure baseline structural properties.
5. Apply each predefined disassembly operator without retuning unrelated representation or structural parameters.
6. Recompute required structural objects under the same frozen measurement procedure.
7. Calculate structural preservation and normalized damage.
8. Generate an integrity profile, response surface, or classification.
9. Optionally perform matched-control comparisons.
10. Only after the integrity analysis is frozen, access downstream outcome data for separate predictive or scientific evaluation if required by the study design.

Methodological separation: A preferred embodiment keeps structural integrity measurement phenotype-blind or target-blind. This makes integrity evidence about the representation itself rather than a disguised downstream-performance metric. Supervised performance may be evaluated later as a separate use case or validation step.

12. Reconstruction and Structural Recovery

Controlled disassembly may be paired with controlled reconstruction. After a component is removed or altered, the system may restore the original component, estimate a replacement, regenerate the component from stored dependencies, or introduce an alternative component. The system then measures the degree of structural recovery.

$$R \quad \text{--disassemble--} \rightarrow \quad R_j \quad \text{--reconstruct--} \rightarrow \quad R_j^{rec}$$
$$Recovery_{(j,k)} = \text{similarity}(M_{k(R)}, M_{k(R_j^{rec})})$$

Reconstruction can distinguish a representation that is fragile but recoverable from one in which disassembly causes persistent structural reorganization. Multiple reconstruction methods may be compared using identical disassembly conditions.

13. Operational Uses of Integrity Measurements

- Select among candidate representations according to structural resilience rather than predictive score alone.
- Trigger local, regional, or global re-indexing when integrity falls below a runtime threshold.
- Request materialization of an additional hierarchy level or structural view when scale dependence is detected.
- Downgrade a representation to stale, degraded, or invalidated runtime state.
- Switch to an alternate representation view whose integrity profile is stronger for the intended operation.
- Request additional data or evidence when integrity is insufficient.
- Identify components that appear structurally indispensable despite low coordinate variance or low individual predictive weight.
- Identify redundant components whose removal produces limited structural damage.
- Guide compression by preserving components with high structural influence per unit mass.
- Provide input evidence to a separate qualification or entitlement subsystem without itself determining scientific entitlement.

13.1 Separation from entitlement

Integrity is preferably treated as a structural measurement rather than a final permission to predict, act, publish, or intervene. A separate evaluation subsystem may determine whether a measured integrity profile is sufficient for a particular scientific or operational purpose. This preserves the distinction between measuring representation behavior and deciding what conclusions the evidence permits.

14. Example PIX / RIX / SII Embodiment

```
PIX / representation runtime
manages views, lifecycle, disassembly jobs, snapshots
```

```

|
RIX / structural computation engine
  computes neighborhoods, recursive regions, paths, transport, stability
|
Integrity subsystem
  compares baseline and controlled-disassembly structure
|
SII / independent qualification
  determines what scientific claims or uses are supported

```

In this embodiment, the integrity subsystem may be implemented as a RIX diagnostic service or as a separate analysis engine. PIX may schedule the controlled modifications and preserve baseline and modified snapshots. SII may consume frozen integrity evidence without changing the underlying representations. These names are illustrative and do not limit the disclosure to a particular product architecture.

15. Experimental Embodiment: Protein Representation Structural Disassembly

One non-limiting experimental embodiment arose from a protein-fitness representation program using an *Escherichia coli* dihydrofolate reductase (DHFR) single-mutant dataset. The integrity analyses described here are structural and phenotype-blind; downstream phenotype labels are not required to compute the structural measurements.

15.1 Baseline direct representation

A high-dimensional descriptor representation was standardized and projected into a 128-component principal-component representation. The baseline representation contained multiple descriptor families, including biochemical, identity, event, alignment-aware, and aggregate features. Structural analysis measured cumulative variance, effective dimension, neighborhood relationships, and edge-distance behavior.

15.2 Family isolation and interaction studies

Developmental studies that isolated selected descriptor families or added explicit pairwise family interactions did not establish a compact predictive mechanism. These negative results motivated investigation of whether useful organization might be distributed across the intact representation rather than attributable to one obvious family or interaction. The present integrity invention does not depend on any predictive conclusion from those studies.

15.3 Phenotype-blind scale disassembly

The direct 128-component representation was subjected to predefined leading-component retention and leading-component removal analyses across multiple q values. Neighborhood preservation and edge-based structural measurements were calculated relative to the intact baseline. The analysis showed that the DHFR representation was not simply globally more diffuse than a comparator protein representation. Instead, local neighborhood organization exhibited scale-dependent sensitivity: leading components were important, while deeper components also became increasingly important as more of the tail was omitted.

A follow-on normalization divided neighborhood damage by the amount of variance removed or omitted. Across the predefined q grid, DHFR exhibited greater variance-normalized sensitivity than the comparator to leading-component removal at all evaluated q values and greater tail-omission sensitivity for a majority of evaluated q values. Edge-rank and tail edge-distance analyses provided additional support, while one head edge-distance measure was discordant. The structural conclusion was therefore expressed as dual neighborhood sensitivity rather than a universal claim about all geometric distances.

15.4 Frozen structural conclusion

The structural branch was frozen as evidence that the DHFR direct representation exhibited greater variance-normalized neighborhood sensitivity to both leading-component removal and trailing-component omission under the predefined analysis. No phenotype labels or predictive models were required for this conclusion. The original held-out DHFR test data remained outside this structural analysis.

Experimental boundary: A later developmental multiscale prediction test was recognized as unsuitable for mechanism inference because of representation-equivalence and scaling confounds. That flawed test is not relied upon here. The integrity disclosure is grounded in the phenotype-blind controlled-disassembly and structural-preservation analyses, not in the flawed downstream mechanism test.

15.5 Illustrative structural values

q	Removed variance DHFR	Removed variance comparator	Tail KNN sensitivity contrast	Head KNN sensitivity contrast	Interpretive use
32	0.311353	0.283162	+0.118322	+0.279460	Both scale directions contribute to local organization
64	0.572603	0.538348	+0.162338	+0.503204	Equal-sized head/tail partition gives strong dual sensitivity
96	0.819802	0.771300	+0.394712	+0.602535	Tail omission becomes especially consequential

These values are included as an enabling example of one implementation and do not define required thresholds, domains, or metrics.

16. Additional Domain Embodiments

World models: Disassemble latent dimensions, path classes, transition edges, or recursive regions and measure whether neighborhoods, trajectories, excursions, or recovery structures persist.

Biological sequence representations: Remove biochemical, identity, conservation, alignment, or structural descriptor families and measure preservation of mutation neighborhoods or transport relationships.

Robotics and autonomous systems: Remove sensor channels, state factors, transition classes, or map levels and measure preservation of reachable operating regimes.

Financial representations: Remove factor blocks, time scales, edge classes, or regime features and measure preservation of market-state neighborhoods and transition topology.

Industrial process models: Remove sensors, feature groups, operating-state dimensions, or process pathways and measure whether known structural regimes remain coherent.

Scientific simulations: Coarsen dimensions, interactions, or multiscale levels and measure which organizational structures are preserved.

Knowledge and reasoning systems: Remove relation types, reasoning steps, evidence families, or graph regions and measure preservation of reachable conclusions or structural explanation paths.

17. Detailed Algorithmic Embodiments

17.1 Algorithm A — Controlled disassembly sweep

```
INPUT: baseline representation R, operators D[1..J], structural metrics M[1..K]
1. Freeze baseline construction parameters.
2. Compute baseline structural measurements  $B_k = M_k(R)$ .
3. For each operator  $D_j$ :
    a. construct modified representation  $R_j = D_j(R)$ 
    b. compute modified structural measurements  $A_{jk} = M_k(R_j)$ 
    c. compute damage  $\Delta_{jk} = d_k(B_k, A_{jk})$ 
    d. compute altered mass  $m_j$ 
    e. compute normalized influence  $S_{jk} = \text{normalize}(\Delta_{jk}, m_j)$ 
4. Assemble integrity profile  $I(R)$ .
5. Store operator, parameter, baseline, modified, and result provenance.
OUTPUT: integrity profile and controlled-disassembly record
```

17.2 Algorithm B — Distributed-dependence detection

```
Apply complementary KEEP(q) and REMOVE(q) operators across predefined q values.
For each q:
    compute local-neighborhood preservation
    compute path/transport preservation
    normalize by omitted or removed mass
If both leading removal and tail omission produce material normalized damage
across predefined scales, classify representation as distributed or multiscale
sensitive
under the selected criterion.
```

17.3 Algorithm C — Integrity-guided compression

```
candidate_components = ordered components or blocks
for c in candidate_components:
    measure structural damage from removing c
    measure representational mass saved
    compute damage_per_saved_mass
select removable components with low structural damage subject to preservation
constraints
rebuild compressed representation
verify structural preservation against baseline
```

17.4 Algorithm D — Reconstruction recovery

```
 $R_j = \text{disassemble}(R, \text{component } c)$ 
for reconstruction method  $A_i$ :
     $R_{rec_i} = \text{reconstruct}(R_j, A_i)$ 
     $\text{recovery}_i = \text{CompareStructure}(R, R_{rec_i})$ 
select reconstruction or alternate representation according to recovery profile
```

18. Controls, Reproducibility, and Scientific Discipline

- Freeze disassembly operator definitions before examining downstream outcomes when target-blind evidence is desired.
- Use matched dimensionality or complexity controls where dimensionality itself could alter structural measurements.
- Retain baseline and modified representation snapshots with full provenance.
- Use deterministic seeds or record stochastic state where randomization is part of the operator.
- Apply paired bootstrap or other resampling when comparing preservation metrics across the same observations.
- Separate exploratory tuning from confirmatory testing and record when a validation set has been reused.
- Do not infer scientific validity, causality, or predictive entitlement solely from an integrity score.
- Document discordant structural measurements rather than collapsing them into a favorable composite without predefinition.

19. Alternative Embodiments and Variations

- The baseline representation may itself be recursive, graph-based, coordinate-based, symbolic, probabilistic, topological, or hybrid.
- Disassembly may be applied once, sequentially, recursively, adaptively, randomly, or according to a predetermined experimental design.
- Operators may alter one component at a time or combinations of components to evaluate redundancy, synergy, or interaction.
- Structural measurements may be exact, approximate, sampled, streaming, local, regional, or global.
- Representational mass may be scalar or multidimensional and may include compute cost or memory cost in addition to mathematical content.
- Integrity profiles may be compared across datasets, time periods, systems, domains, or representation engines.
- An integrity threshold may be absolute, relative to a baseline, relative to a control representation, or learned from historical stable operation.
- The system may abstain from producing a scalar integrity value when different metrics disagree and instead preserve the multidimensional profile.
- Integrity may be computed online for selected structural regions or offline on frozen snapshots.
- Controlled disassembly may be performed physically on stored objects or logically through masked computation without modifying the original representation.

20. Technical Advantages

- Measures intrinsic representation organization without requiring supervised outcome labels.
- Detects structurally important components that may appear minor under variance or coordinate-centric analyses.
- Distinguishes localized, redundant, concentrated, distributed, and multiscale structural dependence.
- Provides a principled basis for compression that preserves organizational structure.
- Supports representation comparison using structural resilience rather than only task performance.
- Can identify hidden fragility before a downstream task visibly fails.
- Generates reproducible evidence through frozen baseline, operator, and modified-representation records.
- Complements existing robustness, ablation, and sensitivity methods while focusing on the representation as the primary object.

21. Illustrative Claim-Style Invention Statements

1. A computer-implemented method comprising constructing or receiving a baseline computational representation; measuring a baseline structural property; applying a controlled disassembly operator to modify fewer than all components of the baseline representation; measuring the structural property in the modified representation; and determining an integrity measure from a difference between the baseline and modified structural properties.
2. The method above wherein the controlled disassembly operator removes, masks, suppresses, shuffles, substitutes, compresses, orthogonalizes, residualizes, isolates, perturbs, or reconstructs one or more representation components.
3. The method above wherein the component comprises a coordinate dimension, principal component, latent factor, feature family, edge class, path class, neighborhood, region, basin, hierarchy level, recursive scale, or generator component.
4. The method above wherein the structural property comprises neighborhood preservation, rank preservation, connectivity, reachability, basin agreement, path preservation, transport preservation, bottleneck preservation, recursive consistency, scale stability, or regime preservation.
5. A method further comprising computing an amount of representational mass altered by the controlled disassembly and normalizing structural damage by the amount of altered representational mass.
6. The method above wherein representational mass comprises explained variance, dimension count, vector norm, feature energy, edge mass, path mass, occupancy, information measure, memory burden, or computational cost.
7. A method comprising applying complementary retention and removal operators across a predefined sequence of component scales and determining whether the representation exhibits distributed structural dependence from structural damage observed in both operator families.
8. A method wherein a component having lower coordinate variance than another component is identified as having greater structural influence per unit representational mass.
9. A method comprising shuffling a representation component while preserving one or more marginal properties and comparing structural damage caused by the shuffled component with structural damage caused by an unshuffled component.
10. A method comprising removing a representation component, reconstructing the component using a reconstruction process, and determining structural recovery from similarity between the baseline representation structure and reconstructed representation structure.
11. A representation-integrity system comprising a disassembly controller, a structural measurement engine, a normalization engine, and an integrity-profile generator.
12. The system above wherein the structural measurement engine is a recursive indexing engine.
13. A method wherein the integrity analysis is performed without access to a downstream supervised target or outcome label.
14. A method further comprising freezing a baseline representation snapshot and a modified representation snapshot for independent replay.
15. A method further comprising selecting between candidate representations based at least in part on their respective integrity profiles.
16. A method further comprising triggering re-indexing, representation replacement, hierarchy materialization, or additional evidence collection when an integrity condition is not satisfied.

17. A method further comprising preserving a multidimensional integrity vector when different structural measurements produce discordant results rather than collapsing the measurements into a single scalar.
18. A method comprising detecting redundancy by identifying a representation component whose removal produces structural damage below a predefined threshold while saving representational mass above a predefined threshold.
19. A method comprising detecting fragility by identifying a representation component whose removal produces structural damage disproportionate to the amount of representational mass removed.
20. A computer-readable medium storing instructions that cause one or more processors to perform any method described herein.
21. A distributed system configured to perform controlled disassembly on selected representation partitions while measuring cross-partition structural preservation.
22. A method comprising comparing integrity profiles of representations derived from different systems or domains using a common structural preservation metric.
23. A method comprising using integrity evidence as an input to a separate qualification or entitlement subsystem without the integrity computation itself conferring entitlement.

22. Relationship to Existing Portfolio and New Matter

Earlier portfolio work relating to recursive indexing, constraint-preserving representation, computational entropy, coordinate-structure divergence, structural excursion/recovery, and structural entitlement provides useful background and may supply structural measurements or operational embodiments for the present disclosure. The July internal memoranda also preserved concepts of representation improvement, multiscale structure, recursive stability, and structural support.

The present disclosure focuses on subject matter believed to be newer and more specific: controlled disassembly of a representation as an integrity experiment; structural remeasurement after disassembly; normalization of structural damage by altered representational mass; multiscale complementary retention/removal protocols; reconstruction-based recovery; and classification of concentrated, redundant, distributed, or fragile structural dependence.

Appendix A — Draft Figure Schematics

FIG. 1

Overall integrity-analysis architecture from baseline representation through controlled disassembly, structural remeasurement, normalization, and integrity profile generation.

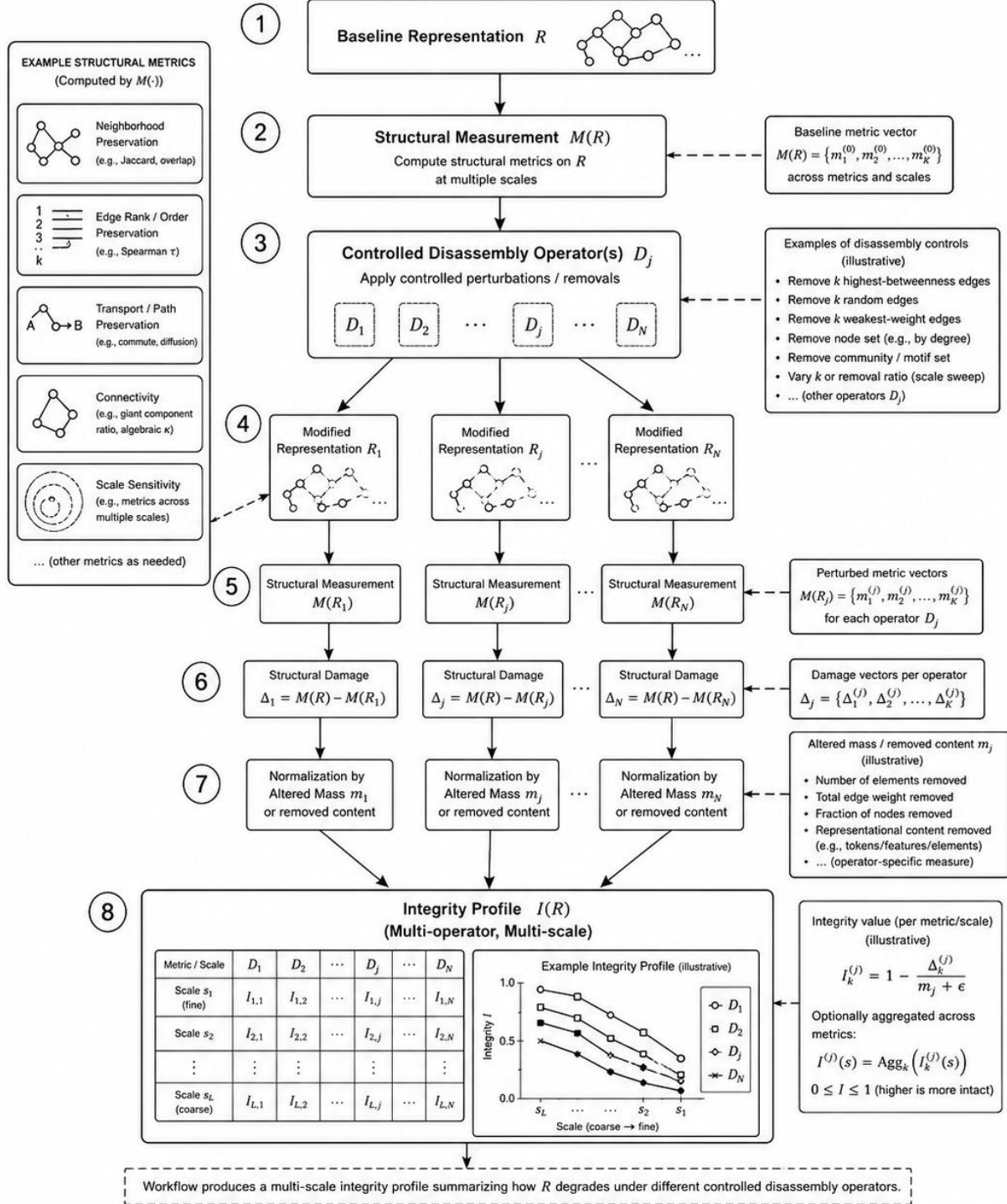


FIG. 1. Overall integrity-analysis architecture from baseline representation through controlled disassembly, structural remeasurement, normalization, and integrity profile generation.

FIG. 2

Example taxonomy of controllable representation components including dimensions, feature families, relationships, paths, regions, scales, hierarchy levels, and generators.

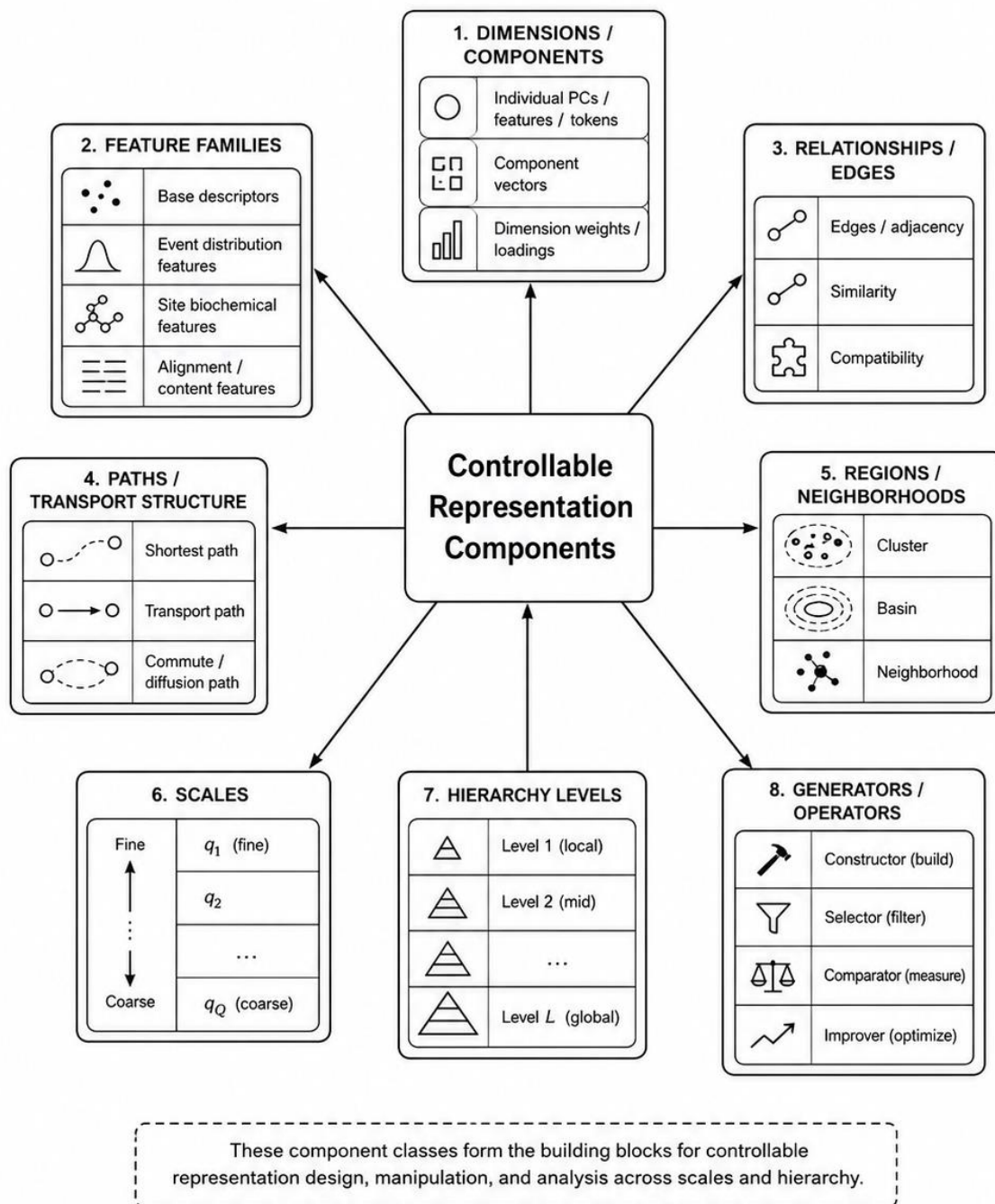


FIG. 2. Example taxonomy of controllable representation components including dimensions, feature families, relationships, paths, regions, scales, hierarchy levels, and generators.

FIG. 3

Example controlled-disassembly operators including removal, masking, shuffling, substitution, compression, orthogonalization, residualization, isolation, and reconstruction.

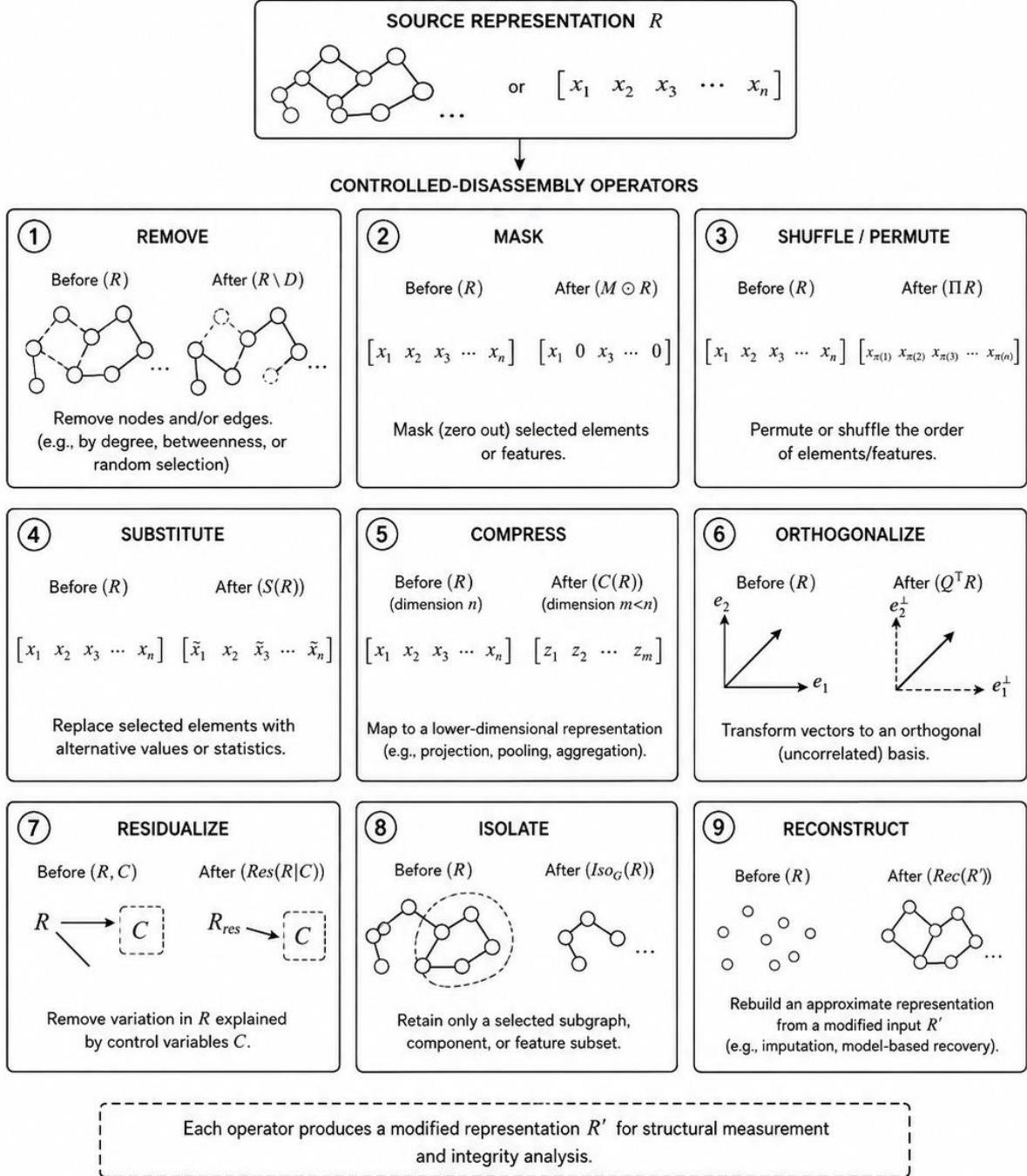


FIG. 3. Example controlled-disassembly operators including removal, masking, shuffling, substitution, compression, orthogonalization, residualization, isolation, and reconstruction.

FIG. 4

Example baseline-versus-modified neighborhood comparison.

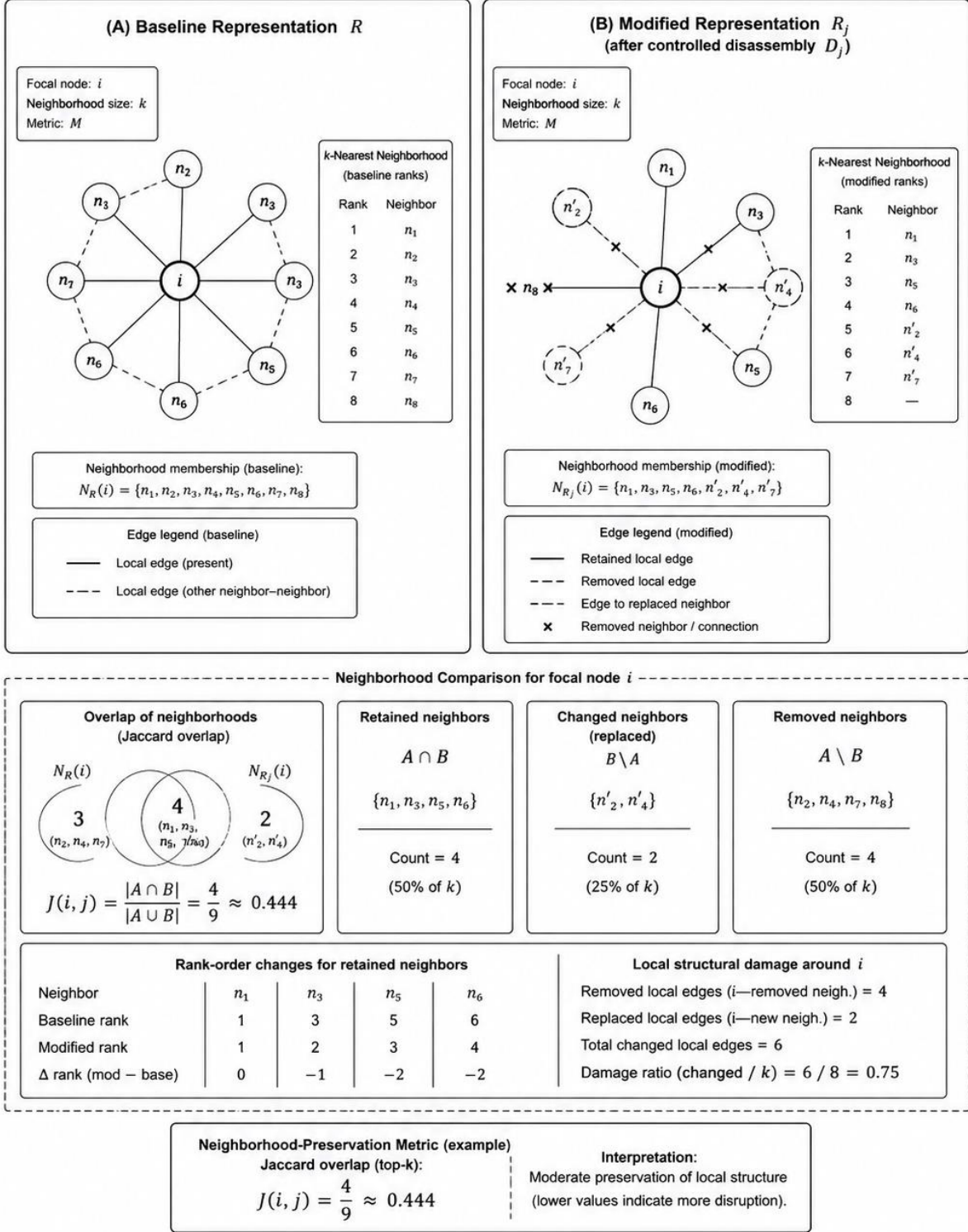


FIG. 4. Example baseline-versus-modified neighborhood comparison.

FIG. 5

Example transport and path preservation analysis.

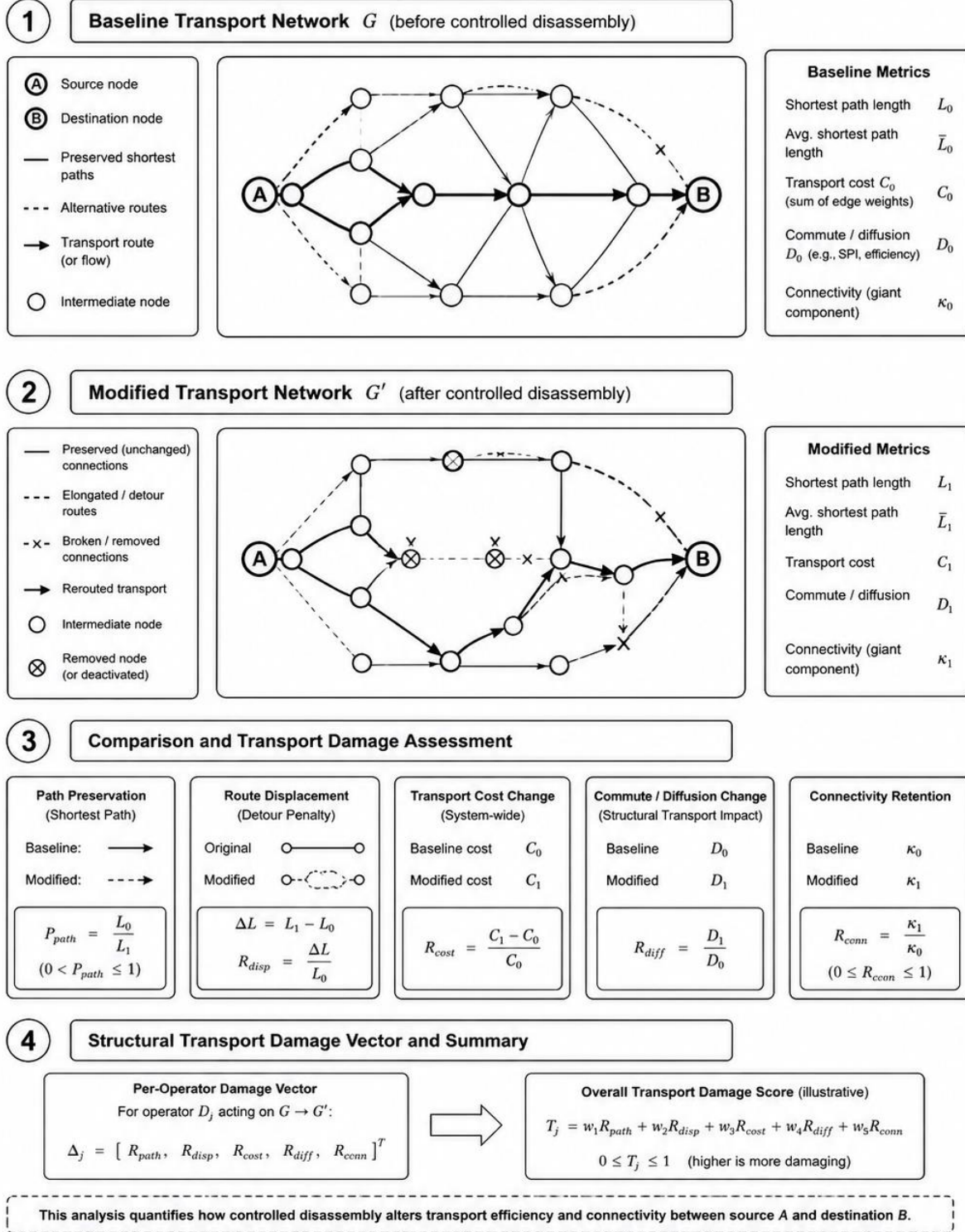


FIG. 5. Example transport and path preservation analysis.

FIG. 6

Example normalization of structural damage by removed representational content.

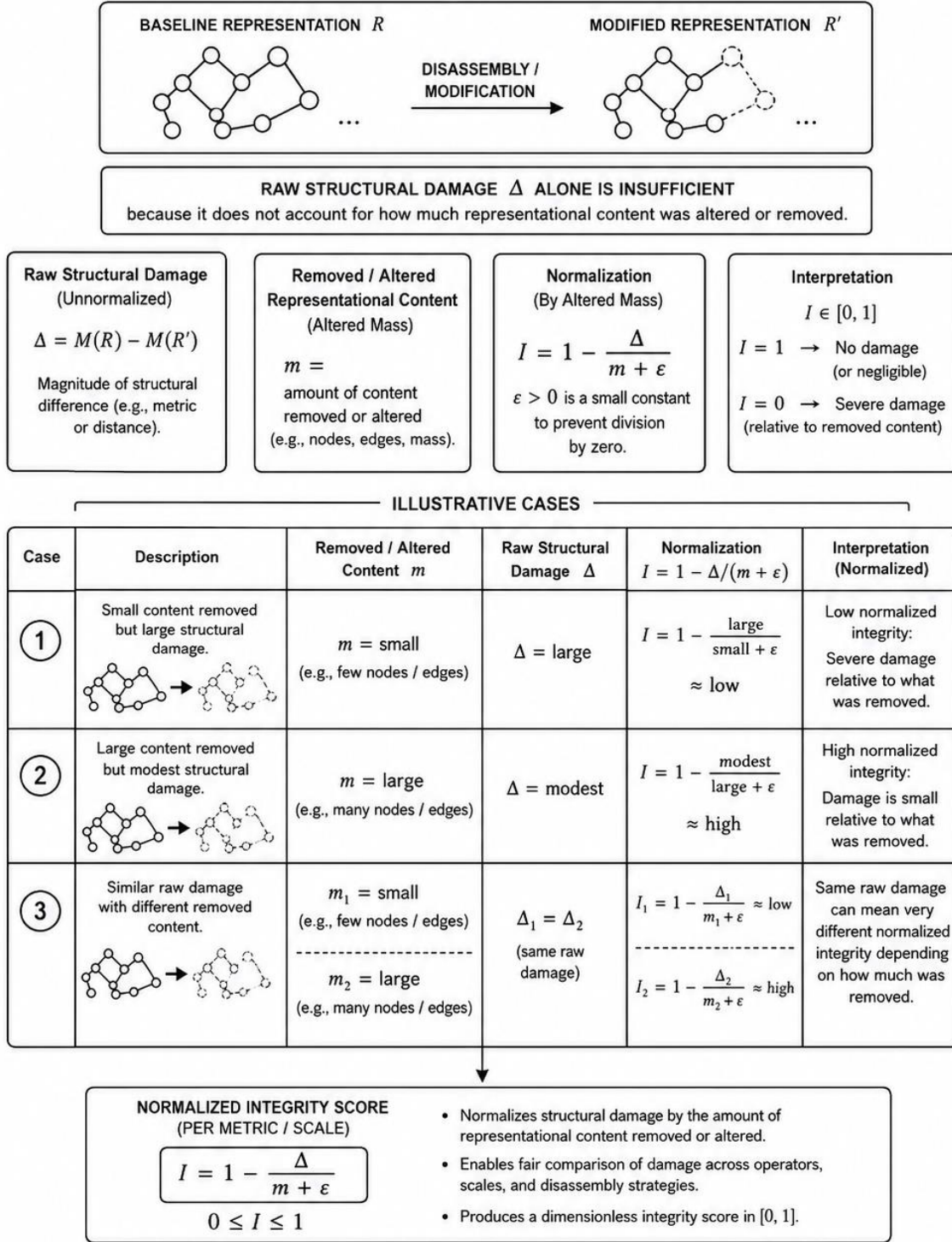


FIG. 6. Example normalization of structural damage by removed representational content.

FIG. 7

Example integrity profile across multiple disassembly scales.

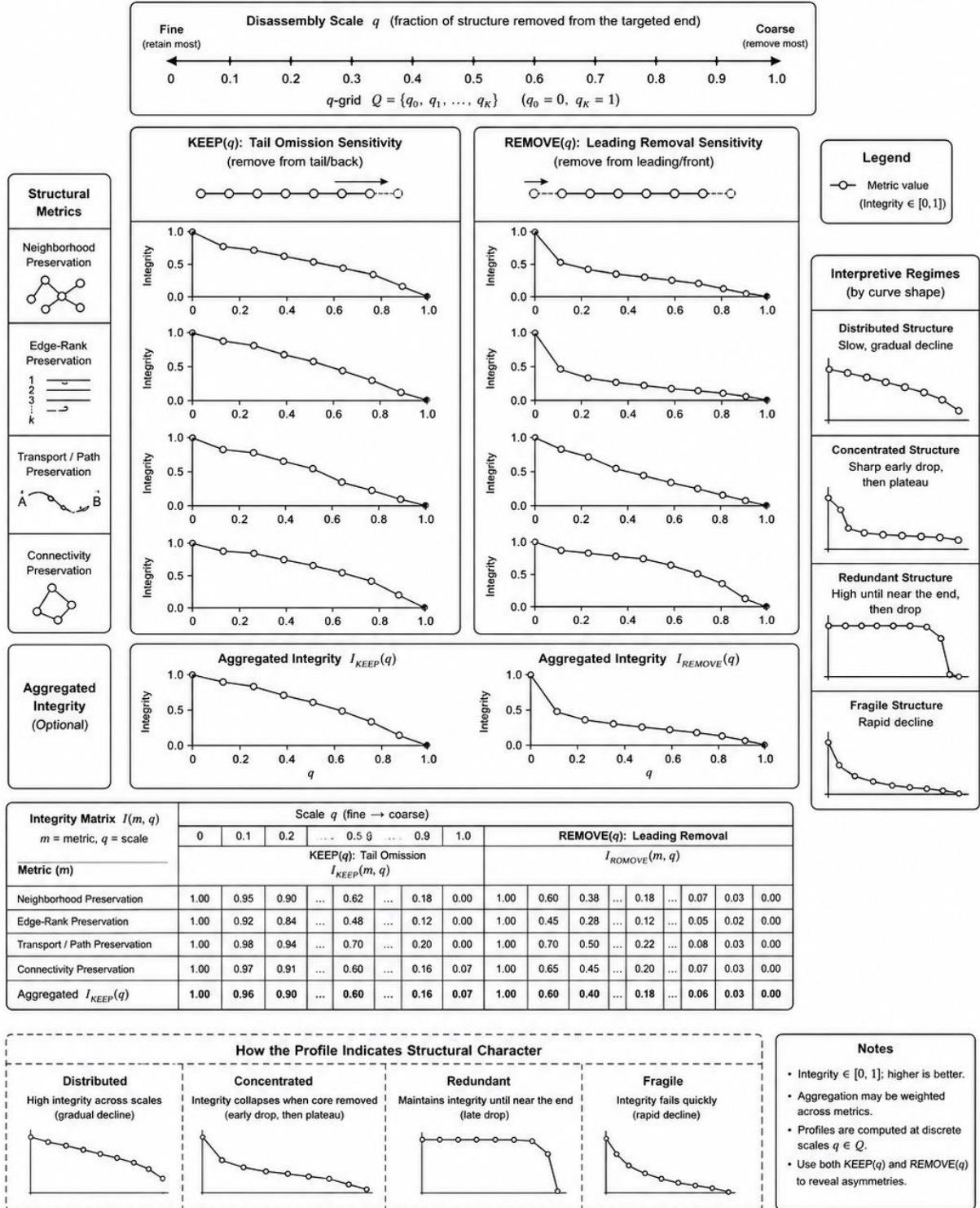


FIG. 7. Example integrity profile across multiple disassembly scales.

FIG. 8

Example distributed-structure condition in which both leading and trailing representation components contribute to local organization.

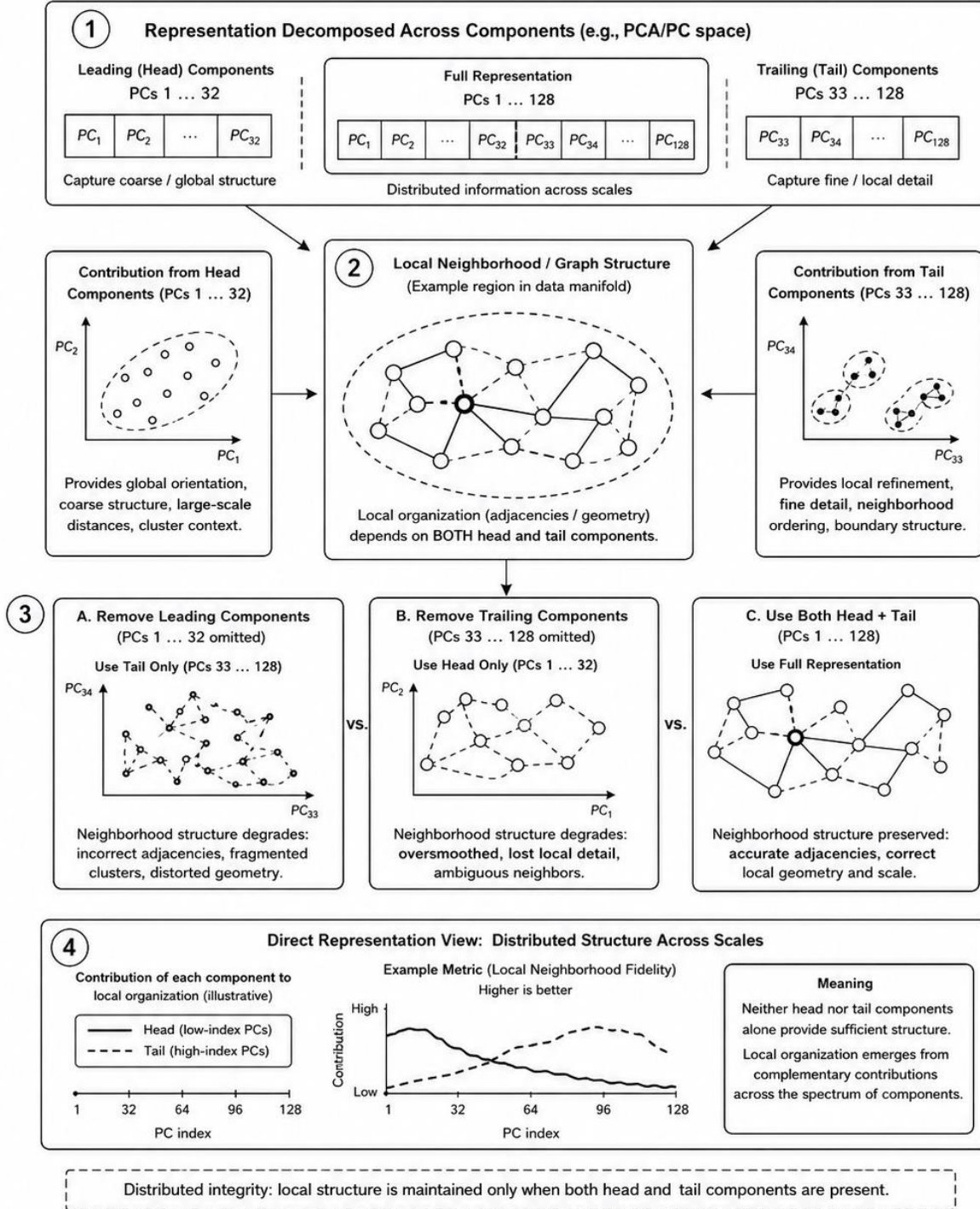


FIG. 8. Example distributed-structure condition in which both leading and trailing representation components contribute to local organization.

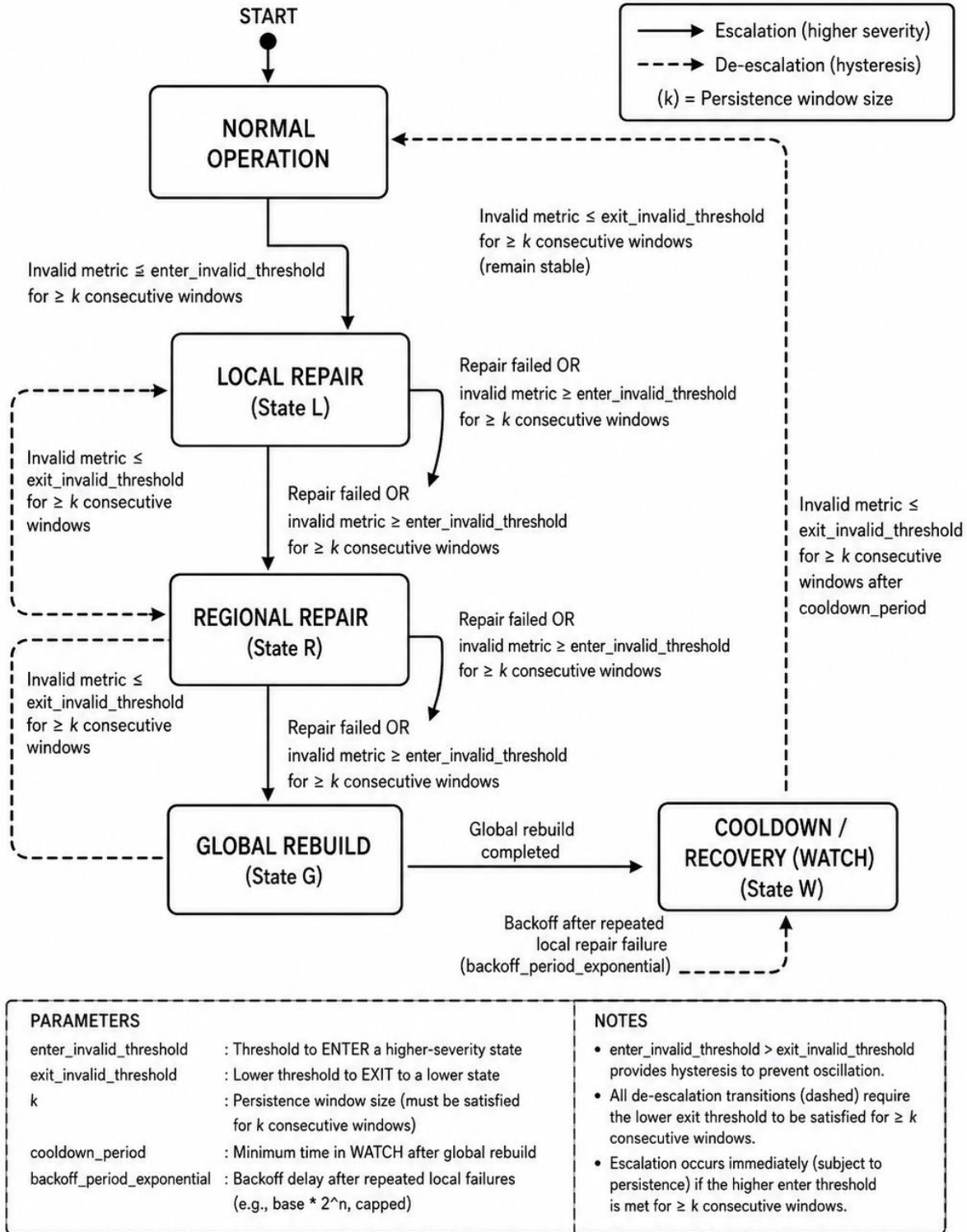


FIG. 9

FIG. 9. Example reconstruction/recovery integrity test.

FIG. 10

Example interaction with a representation runtime, structural engine, and independent qualification subsystem.

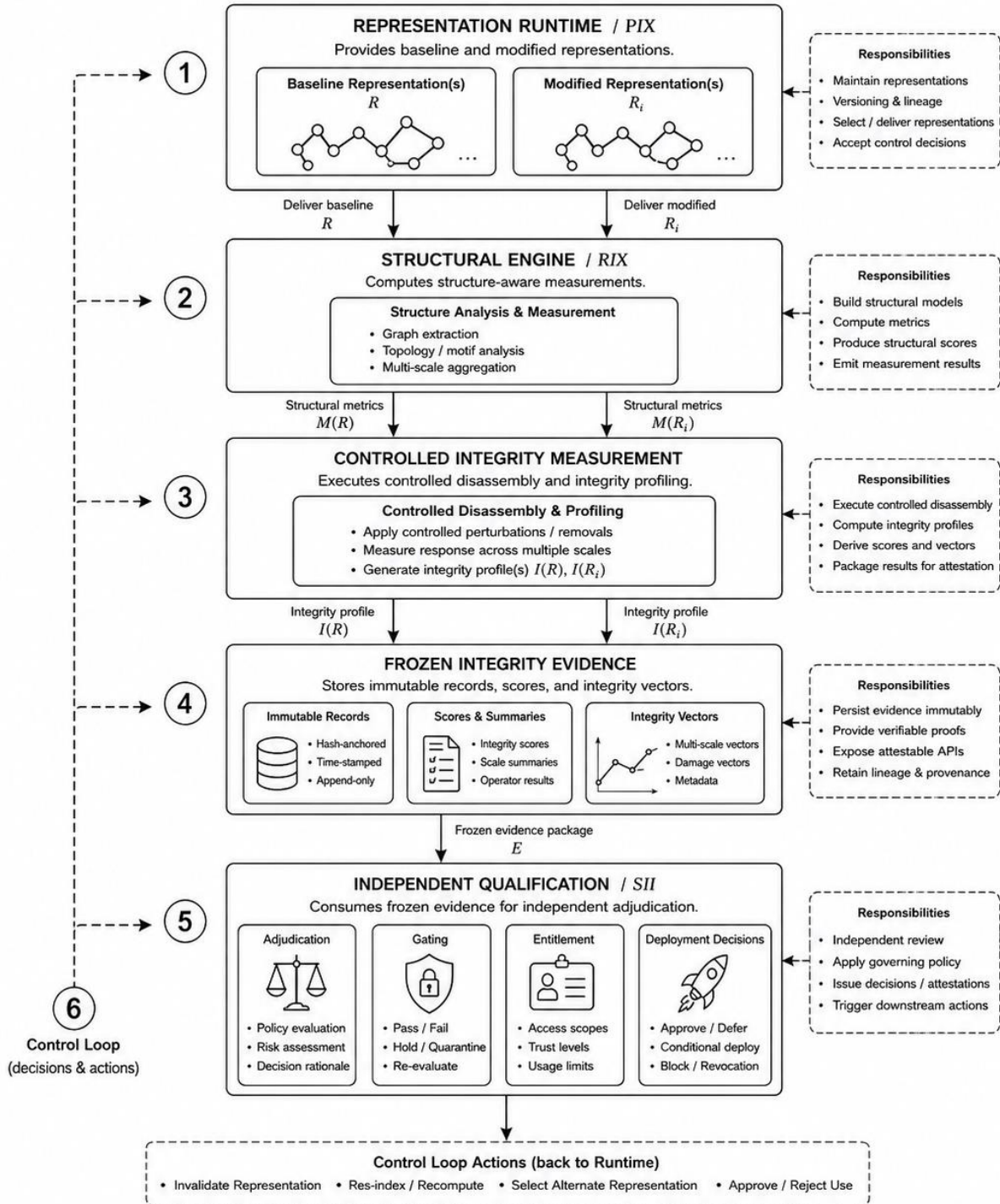


FIG. 10. Example interaction with a representation runtime, structural engine, and independent qualification subsystem.

FIG. 11
Phenotype-Blind Experimental Protocol Separating Representation Integrity from Later Outcome Evaluation

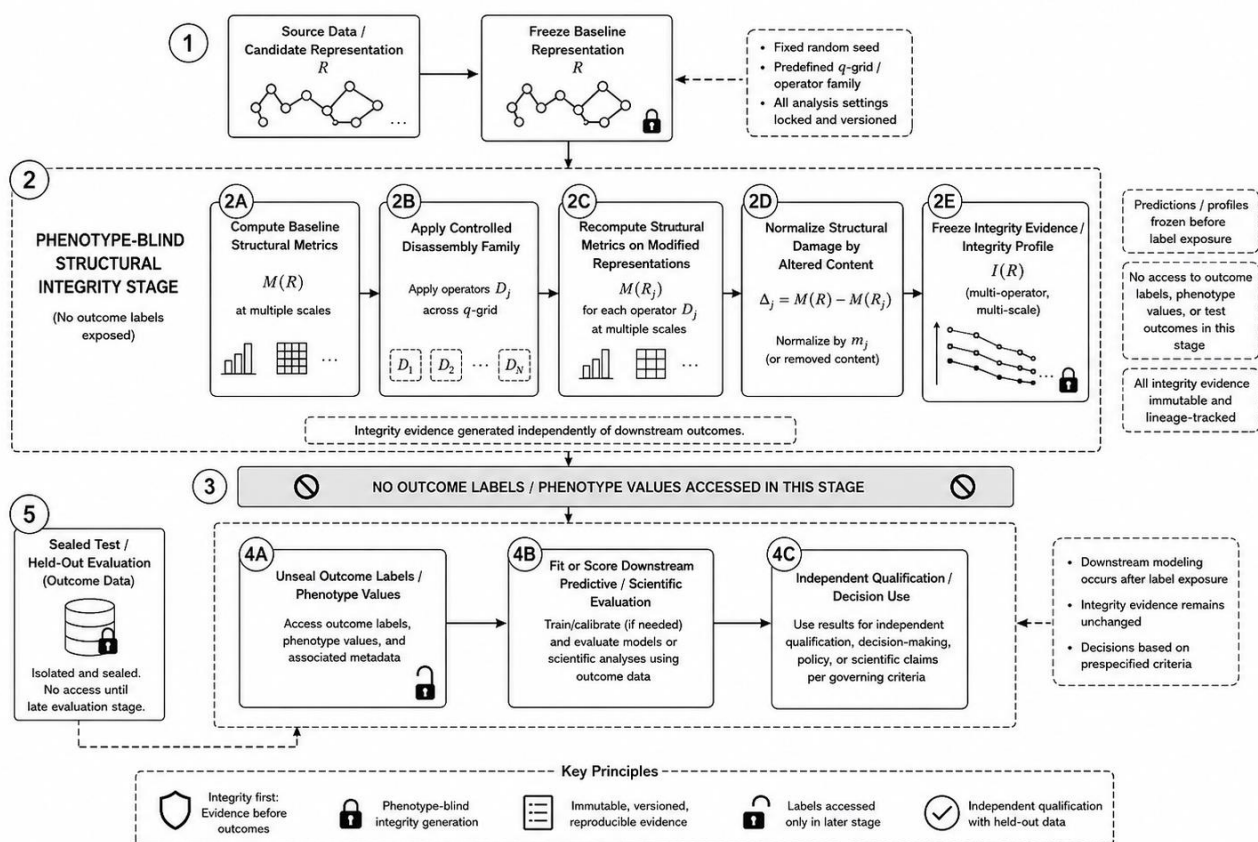


FIG. 11. Example phenotype-blind experimental protocol separating representation integrity from later outcome evaluation.

FIG. 12

Cross-Representation Comparison Using Matched Disassembly Operators

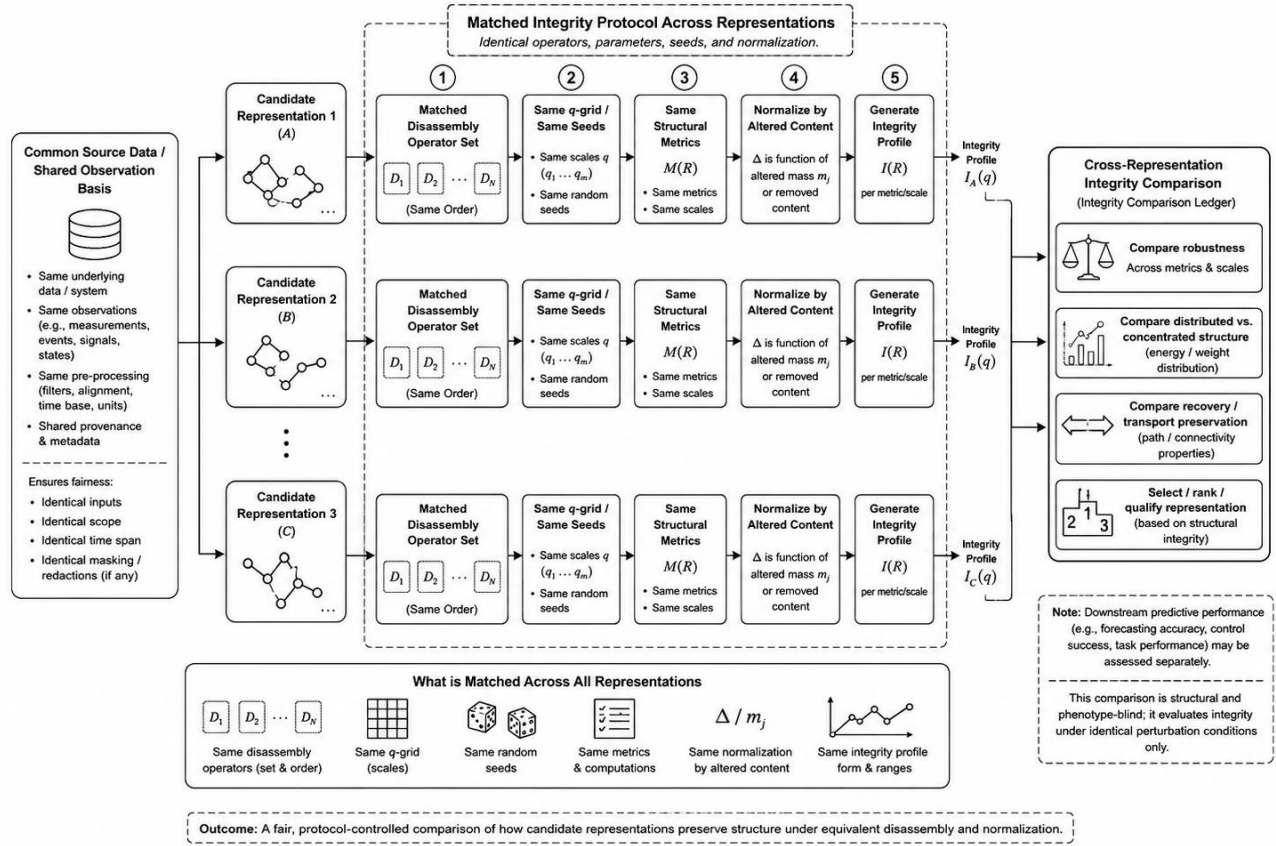


FIG. 12. Example cross-representation comparison using matched disassembly operators.

Appendix B — Example Phenotype-Blind Protocol Schema

```

IntegrityProtocol {
  baseline_representation: frozen
  structural_metrics: [knn_overlap, edge_rank, transport, connectivity]
  disassembly_family: [KEEP(q), REMOVE(q)]
  q_grid: predefined
  normalization: variance_removed_or_omitted
  supervised_targets_accessed: false
  downstream_test_data_accessed: false
  random_seed: fixed
  result_state: immutable / frozen
}

```